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STAFF REPORT FOR THE 2014 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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This debt sustainability analysis builds on the one conducted at the time of the 2013 Article IV Consultation. It finds that Cameroon's external debt remains sustainable, with all external debt ratios staying below their respective thresholds. However, the risk of debt distress has increased from "low" to "moderate," as the external debt indicator for the present value of debt over exports breaches the policy-dependent threshold under a classic shock to exports. The underlying macroeconomic assumptions in this analysis are comparable to those used in the previous one. However, the composition and terms of new external debt have been adjusted to reflect the large, undisbursed debt overhang contracted on nonconcessional terms to finance the country's public investment program. Heavier external borrowing than projected could jeopardize debt sustainability over the long term and calls for a prudent approach regarding the terms of borrowing. Moreover, significant vulnerabilities related to a weak domestic debt policy, and a fiscal situation projected to become increasingly fragile, heighten the risk of overall debt distress.

BACKGROUND

1. This debt sustainability analysis (DSA) of Cameroon's public debt was prepared jointly by the International Monetary Fund (IMF) and the World Bank. It updates the 2013 DSA (IMF Country Report No. 13/279).[1] It uses the latest standard debt dynamic template for low-income countries,[2] based on the most recent external debt data from the authorities, and the macroeconomic framework derived from the 2014 IMF Article IV consultation. Data are composed of external and domestic debt of the central government, as well as debt and guaranteed debt of public enterprises. As noted in the previous DSA, debt statistics could benefit from a more comprehensive coverage of contingent liabilities on financial institutions and liabilities of public enterprises and municipalities.

2. The enhanced Heavily Indebted Poor Countries (HIPC) and the Multilateral Debt Relief Initiatives resulted in a significant decline in Cameroon's public debt ratios, from more than 50 percent of GDP in 2005 to less than 10 percent of GDP in 2008. However, the public debt-to-GDP ratio has been steadily increasing since then, reaching 19.5 percent at end-2013. This increase mainly corresponds to a rise in public external debt, representing 63.4 percent of total public debt at end-2013 (Text Table 1). Bilateral loans from non-Paris club members have significantly grown, representing 30.7 percent of total public debt at end-2013, as opposed to only 4.3 percent at end-2008. This evolution reflects an increasing number of loans, with less favorable financing conditions, to finance the authorities' public investment program. Although there was a fall in structured domestic debt in 2012, the overall stock of domestic debt has remained on a rapid growth path since 2011 (Text Table 2).

Text Table 1. Cameroon: Stock of Public Debt, 2006–13

	2006	2007	2008	2009	2010	2011	2012	2013
				(CFAF billions)				
Total public debt	1,489	1,171	1,015	1,114	1,349	1,662	2,085	2,772
External debt	603	562	578	574	725	927	1,197	1,757
Multilateral	206	230	289	377	460	577	632	901
Bilateral	316	289	288	196	222	304	470	989
Bilateral Paris Club	294	251	245	63	91	85	75	139
Bilateral non-Paris Club	22	38	43	133	132	219	395	851
Commercial debt	81	43	1	1	43	46	96	138
Domestic debt	887	608	437	540	623	734	888	1,014
				(Percent of total)				
Total public debt	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
External debt	40.5	48.0	56.9	51.5	53.8	55.8	57.4	63.4
Multilateral	13.8	19.7	28.5	33.8	34.1	34.8	30.3	32.5
Bilateral	21.2	24.7	28.4	17.6	16.5	18.3	22.6	35.7
Bilateral Paris Club	19.7	21.4	24.2	5.7	6.7	5.1	3.6	5.0
Bilateral non-Paris Club	1.5	3.2	4.3	12.0	9.8	13.2	18.9	30.7
Commercial debt	5.4	3.7	0.1	0.1	3.2	2.8	4.6	5.0
Domestic debt	59.5	52.0	43.1	48.5	46.2	44.2	42.6	36.6
				(Percent of GDP)				
Total public debt	15.9	12.0	9.7	10.1	11.5	13.2	15.6	19.5
External debt	6.4	5.7	5.5	5.2	6.2	7.4	9.0	12.5
Multilateral	2.2	2.4	2.8	3.4	3.9	4.6	4.7	6.4
Bilateral	3.4	2.9	2.8	1.8	1.9	2.4	3.5	7.0
Bilateral Paris Club	3.1	2.6	2.3	0.6	0.8	0.7	0.6	1.0
Bilateral non-Paris Club	0.2	0.4	0.4	1.2	1.1	1.7	2.9	6.0
Commercial debt	0.9	0.4	0.0	0.0	0.4	0.4	0.7	1.1
Domestic debt	9.4	6.2	4.2	4.9	5.3	5.9	6.6	7.0
Memorandum item:								
Nominal GDP	9,388	9,792	10,444	11,040	11,700	12,546	13,515	14,463

Sources: Cameroonian authorities; and IMF and World Bank staff estimates.

[1] The draft DSA was discussed with the Cameroonian authorities in the course of the 2014 Article IV consultation. This DSA follows the IMF and World Bank Staff Guidance Note on the Application of the Joint IMF-World Bank Debt Sustainability Framework for Low-Income Countries, dated November 7, 2013 (SM/13/292).

[2] The template was provided by the IMF's Strategy and Policy Review Department.

Text Table 2. Cameroon: Domestic Debt Components, 2011–13
(CFAF billions; unless otherwise indicated)

	2011	2012	2013	2013 Share in percent
Total domestic debt	734	888	1,014	100
Structured debt	571	498	515	51
Banking	123	117	108	11
Non-banking	448	381	407	40
Non structured debt[1]	163	390	499	49

Sources: Cameroonian authorities; and IMF and World Bank staff estimates.
[1] Includes financing gap.

UNDERLYING ASSUMPTIONS

3. The new macroeconomic framework reflects limited adjustments to projections of key variables compared to the 2013 DSA framework. Oil revenue is somewhat lower, and growth and exports are assumed to be somewhat less buoyant in the medium term, but more resilient over the long term (Text Table 3 and Box 1). Oil revenue, as a percent of GDP, is expected to fall because increases in oil production are offset by a projected fall in international oil prices and rising production costs. Real GDP growth remained strong in 2013, thanks to better implementation of the public investment plan, and is expected to remain robust in the medium term, based on sustained domestic demand and the coming onstream of infrastructure projects. The volume of non-oil exports is projected to grow by around 9 percent a year, above real GDP growth. The volume of imports is projected to grow by around 7 percent per year, at a somewhat higher pace than GDP. Inflation is projected to remain well below the 3 percent convergence criterion of the Central African Economic and Monetary Community (CEMAC).

Text Table 3. Cameroon: Key Macroeconomic Assumptions, 2013–34

	2013-14[1]	2015-19[2]	2020-34[3]
Real GDP growth (percent)			
DSA 2014	5.5	5.5	5.5
DSA 2013	4.6	5.1	4.8
Total revenue (percent of GDP)[4]			
DSA 2014	18.3	17.7	16.4
DSA 2013	18.9	19.3	18.3
Exports of goods and services (percent of GDP)			
DSA 2014	27.1	25.5	24.2
DSA 2013	28.7	27.4	22.9
Oil price (US dollars per barrel)			
DSA 2014	108.0	93.3	91.7
DSA 2013	106.0	88.5	93.2

Sources: Cameroonian authorities; and IMF and World Bank staff estimates.
[1] 2013 DSA referred to periods 2012-13.
[2] 2013 DSA referred to periods 2014-17.
[3] 2013 DSA referred to periods 2018-33.
[4] Total revenue, excluding grants.

4. However, some assumptions have been adjusted. The discount rate increased from 3 percent for the 2013 DSA to 5 percent, as approved by the IMF and World Bank Executive Boards in October 2013. The level of external public debt at end-2012 was adjusted upward to CFAF 1,197 billion from CFAF 1,127 billion, and again to CFAF 1,595 billion at end-2013. Finally, given new information of large as-yet-undisbursed debt contracted at nonconcessional rates (over CFAF 2,700 billion at end-2013 versus about CFAF 2,100 billion at end 2012; Text Table 4), the composition of debt by level of concessionality from 2013 forward was revised (Text Table 5). The DSA assigns a larger role to non-Paris Club bilateral creditors, whose loans are nonconcessional, especially in 2015 and 2016, given specific information on undisbursed loans. Their share of new external financing rises from 49.5 percent of all new debt accrued at end-2014, reaches a maximum of 51.2 percent by 2015, and falls to 36.9 percent from 2017 onward; the composition of new debt in the 2013 DSA was characterized by a slower increase in the share of non-Paris Club members (from 28.8 percent of the total in 2012 to 41.9 percent by end-2018).

5. Other important assumptions for the DSA remain unchanged, including the assumption of a limited overall amount of future external debt. The fiscal financing gap is assumed to lead to an accumulation of mostly domestic debt, which does not come into play when assessing the risk of external debt distress. Some contingent liabilities in state-owned enterprises (SOEs) have been left out, because although they may be sizable, their overall amount remains hard to quantify.

Box 1. Cameroon: Macroeconomic Assumptions for the Baseline Scenario[1]

Medium Term, 2015–19

- Real GDP growth is projected to reach an average of 5.5 percent in the medium term, supported by a strong domestic demand, a rapid increase in oil production, and high public investments. Annual inflation is projected to remain low, at 2.2 percent, in line with historical trends and below the CEMAC convergence criterion.
- The revenue-to-GDP ratio is projected to dip slightly over the medium term, from 17.7 percent of GDP in 2015 to 17.3 percent of GDP in 2019. Although oil prices are expected to ebb over the medium term, production of oil is assumed to be higher thanks to the recent discovery of an oil field and new technology to tap old wells. However, this technology is more expensive and will reduce oil revenue in proportion.
- The external current account deficit is projected to grow from 4.2 percent of GDP in 2015 to 4.4 percent of GDP in 2019. The deficit is assumed to be driven by more imports, which in turn stem from real GDP growth and the need for imported equipment and intermediate goods for infrastructure projects. As before, the current account deficit is expected to be financed through foreign direct investment, external public borrowing, and other private capital inflows.

Long Term, 2020–34

- Real GDP growth is projected to average 5.5 percent. Long-term growth is driven by non-oil exports, domestic demand, and a sustained rate of private capital spending consistent with the advent of a more favorable business climate.
- The revenue-to-GDP ratio is projected to decrease from 17.3 percent of GDP in 2020 to 15.3 percent of GDP in 2034. This trend assumes oil revenue will be decreasing with the depletion of oil reserves, while non-oil revenue is still sustained by further structural reforms to improve revenue collection and to usher in a more diversified economy.
- The external current account deficit is projected to decline gradually to 3.5 percent of GDP in 2034, on account of a strong non-oil export growth facilitated by the coming onstream of new infrastructure, such as the Kribi deep-sea port, while oil exports become less important.

[1] The baseline scenario uses the latest IMF World Economic Outlook assumptions (March 2014).

Text Table 4. Cameroon: Committed but Non-Disbursed Loan Stocks by Category of Creditors, 2010–13
(CFAF billions)

	2010	2011	2012	2013
Multilateral	613	630	727	826
Bilateral	156	697	1,286	1,807
Commercial	...	94	83	85
Total	769	1,422	2,095	2,718

Source: Cameroonian authorities.

Text Table 5. Cameroon: Allocation of New External Commitments by Concessionality Levels, 2013–34
(CFAF billions)

	2013	2014	2017	2020	2025	2030	2034	Average 2015-19[1]	Average 2020-34[2]
New borrowing, 2013 DSA	265	348	462	491	559	616	...	394	564
Concessional	59	72	80	75	77	77	...	73	76
Percent of total	22	21	17	15	14	12	...	19	14
Nonconcessional	207	276	382	417	482	539	...	321	488
Percent of total	78	79	83	85	86	88	...	81	86
New borrowing, 2014 DSA[3]	487	574	463	520	728	985	1,273	495	852
Concessional	147	170	211	234	321	423	538	201	370
Percent of total	30	30	46	45	44	43	42	41	43
Nonconcessional	340	404	252	286	407	561	734	295	482
Percent of total	70	70	54	55	56	57	58	59	56

Sources: IMF and World Bank staff estimates and projections.

[1] 2013 DSA referred to periods 2013-18.

[2] 2013 DSA referred to periods 2019-33.

[3] Includes external borrowing by SOEs of 7 percent of public external borrowing in 2016 and 2017, peaking at 9 percent in 2028.

EXTERNAL DEBT SUSTAINABILITY ASSESSMENT

6. The Low Income Countries (LICs) debt sustainability framework is guided by country-specific indicative debt burden thresholds for external debt. These thresholds reflect the empirical findings that sustainable debt levels for a LIC increase with the quality of policies and institutions, which is measured by the World Bank's Country Policy and Institutional Assessment (CPIA) index (Text Table 6). This DSA is informed by Cameroon's CPIA score for 2012. The score of 3.23, on a scale of 1 to 6, has virtually not changed over the last five years and corresponds to an assessment of weak institutional capacity. Cameroon's rank is better than the CEMAC average (2.9) and in line with the sub-Saharan African (SSA) average. The indicative external debt burden thresholds for countries in this category are (i) a present value (PV) of the debt-to-exports ratio of 100 percent; (ii) a PV of the debt-to-revenue ratio of 200 percent; (iii) a PV of the debt-to-GDP ratio of 30 percent; and (iv) debt service-to-exports and revenue ratios of 15 percent and 18 percent, respectively.

Text Table 6. Cameroon: Country Policy and Institutional Assessment Ratings, 2008–12[1]

	2008	2009	2010	2011	2012
Cameroon	3.21	3.21	3.17	3.18	3.23
CEMAC[2]	2.74	2.79	2.80	2.84	2.86
Sub-Saharan Africa[2]	3.15	3.17	3.21	3.20	3.17

Source: World Bank, World Development Indicators (2013).

[1] CPIA ratings measure the quality of a country's policies and institutions; they range from 1 (Low) to 6 (High).

[2] Poverty Reduction and Growth Trust (PRGT) eligible countries.

7. Cameroon's external debt remains sustainable, with all external debt ratios staying below their respective thresholds (Text Table 7 and Figure 1). There is, however, an upward trend for all debt ratio indicators, before subsequently abating at the end of the projection period. This is due to the assumption that the large fiscal financing gaps in the macroeconomic framework are to be financed by domestic debt, while external debt accumulation remains on a more modest path by comparison. Moreover, the grant element of new external borrowing is on average lower than in the previous DSA, while other relevant variables do not change much. These developments lead to a bigger challenge to finance public investments sustainably.

Text Table 7. Cameroon: Baseline Debt Ratios, 2014–34

(Percent)

	Threshold	2014	Medium term 2015-19	Long term 2020-34
	Debt Sustainability Analysis, 2014			
External debt				
PV of debt-to-GDP	30	10.2	12.9	13.5
PV of debt-to-exports	100	37.7	50.6	55.9
PV of debt-to-revenue	200	56.4	74.1	83.1
Debt service-to-exports	15	2.7	3.7	6.2
Debt service-to-revenue	18	4.0	5.5	9.2
Public debt				
PV of debt-to-GDP		19.5	29.0	50.9
PV of debt-to-revenue		105.1	164.3	313.6
Debt service-to-revenue		9.6	14.4	21.3
		2013	2014-18	2019-33
	Debt Sustainability Analysis, 2013			
External debt				
PV of debt-to-GDP	30	7.7	11.1	14.7
PV of debt-to-exports	100	27.0	41.0	64.9
PV of debt-to-revenue	200	39.4	58.0	80.8
Debt service-to-exports	15	1.6	2.2	4.0
Debt service-to-revenue	18	2.4	3.1	5.0
Public debt				
PV of debt-to-GDP		17.2	25.9	44.5
PV of debt-to-revenue		86.5	132.7	244.8
Debt service-to-revenue		7.7	10.5	14.1

Sources: IMF and World Bank staff estimates and projections.

8. However, Cameroon's risk of external debt distress, based on strenuous standard stress testing, has increased from "low" to "moderate." A shock to exports causes the ratio of the PV of external debt to exports to exceed 100 percent by 2016 (Figure 1, panel c). This new development reflects the higher external debt stock at end-2013, the magnitude of the standard stress test,[3] and, to a lesser extent, the change toward less concessionality in the composition of new external debt.

PUBLIC SECTOR DEBT SUSTAINABILITY ASSESSMENT

9. As in the previous DSA, the inclusion of domestic debt exacerbates the pace of accumulation of public debt, and shows a clear deterioration in the debt ratios. The stock of domestic debt is projected to increase faster than external debt over the medium term, as the projected fiscal financing gaps are rolled up into domestic debt, in the form of new government securities, more securitization of arrears, and the accumulation of further domestic arrears. Therefore, the PV of debt-to-GDP ratio is expected to reach higher levels than in the previous DSA, from 19.5 percent of GDP in 2014 to almost 50.9 percent of GDP in 2034, after breaching the indicative ceiling of 38 percent of GDP in 2021. The PV of debt-to-revenue and the PV of debt service-to-revenue show a similar upward trajectory.

10. The overall amount and the uncertain financing terms of the financing gaps, and the uncertain level of contingent liabilities add another element of risk. While Cameroon faces a moderate risk of debt distress based on an assessment of public external debt, it faces a higher overall risk of debt distress reflecting significant vulnerabilities related to its weak domestic debt policy, such as a roll-over risk.

CONCLUSION

11. This DSA confirms the growing risk of an unsustainable accumulation of total public debt. Its mitigation requires a mix of a more conservative fiscal policy and more favorable debt terms. Although Cameroon's current debt burden remains low, both domestic and external borrowings have expanded at a rapid pace and at increasingly nonconcessional terms. Risks are growing because future financing needs remain unidentified and reasonable pricing and maturities may not be secured.

12. On the basis of this DSA, Cameroon's external debt remains sustainable, but its risk of external debt distress has increased from "low" to "moderate." All debt indicators trend below their policy-dependent thresholds, but a shock to exports causes the ratio of PV of external debt to exports to exceed 100 percent by 2016. This underscores Cameroon's particular vulnerability to an oil price shock, or a more limited increase in oil production than currently foreseen. This result calls for the adoption of prudent debt policies favoring more concessional terms, and the importance of economic growth to help diversify the export base and generate additional revenue. Nonconcessional loans should only be considered for well-assessed, high-yield commercial or infrastructure projects that will generate sufficient government

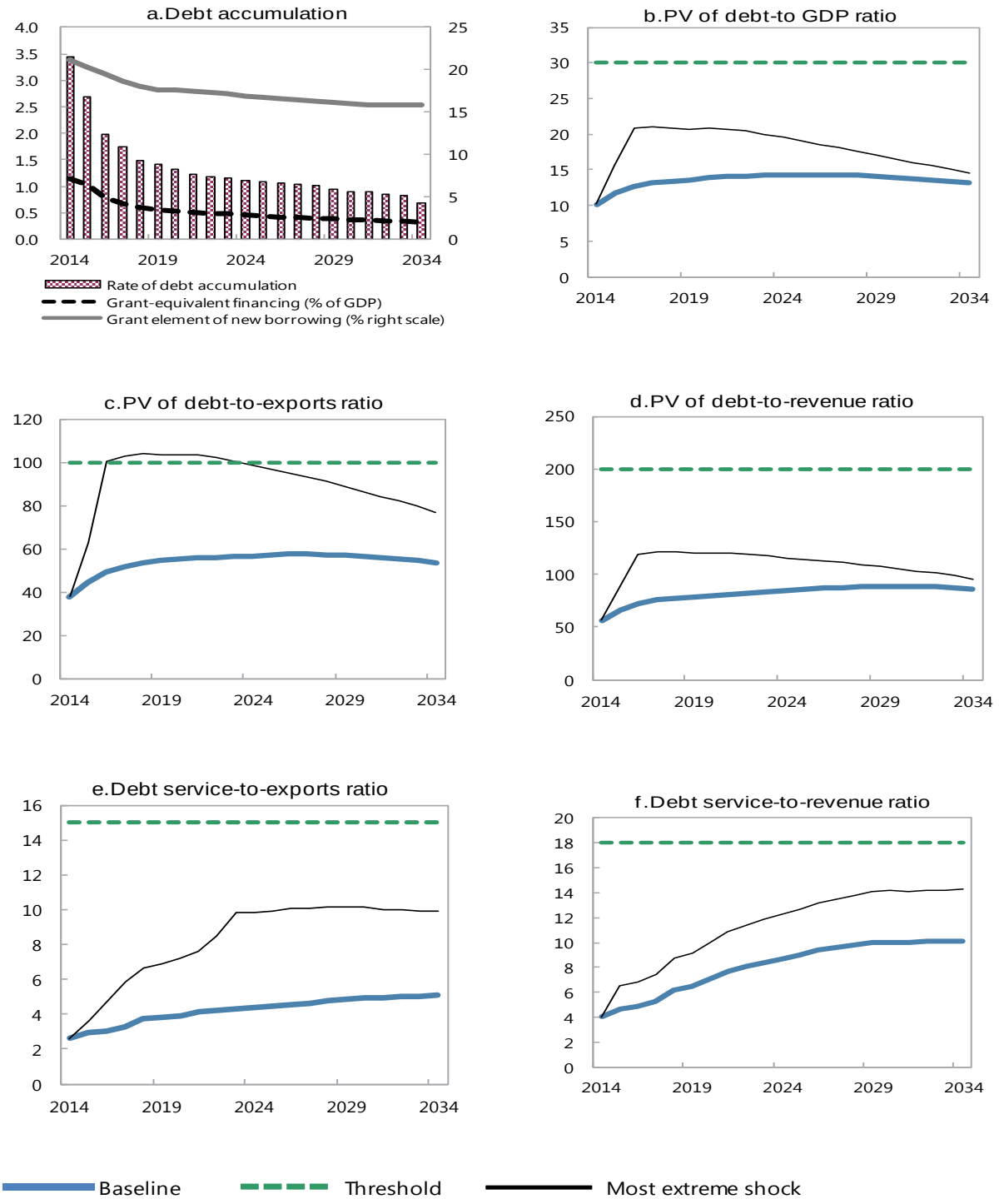
[3] The relevant "most extreme" stress test is a growth rate of exports at its 10-year historical average minus one standard deviation applied over 2015 and 2016, resulting in a negative growth of exports of 7.1 percent each year (versus a negative growth of 5.7 percent for two successive years in the 2013 DSA). A shock resulting in negative growth of exports of less than 5.8 percent for 2015 and 2016 would not cause a breach of the policy-dependent threshold, nor would any of the other standard stress tests.

revenue to cover debt service related to the projects. Export growth will originate primarily from the private sector, which requires a more favorable business climate.

13. The rapid increase in external debt between the previous and this DSA points to significant risks related to insufficient data coverage, lags in communication with some creditors, and, to date, inoperative upstream controls on new debt contracting. Current project implementation arrangements allow for direct payments between foreign contractor and external lender, which need to be reported without delay to the Government. The creation of the National Public Debt Committee (NPDC) has been a step in the right direction, but questions remain as to its effective role in the process of evaluating and contracting new debt. Given the rapid rise in external debt and the large undisbursed debt overhang, staffs recommend subjecting new disbursements of external debt to a prior approval by the NPDC.

14. The Cameroonian authorities have indicated their agreement with the analysis and conclusions reached in this DSA. They agreed that it is essential to maintain debt sustainability, and share the view that current debt trends, if left unaddressed, pose a risk to the country's long-term fiscal stability. Staffs recommended to subject new loans to a review by the NPDC to ascertain their consistency with debt sustainability. The authorities noted that given the large financing needs in the near term, they may have to contract nonconcessional loans in the absence of available alternative concessional financing for projects with a high return. The authorities are working with the World Bank regarding the ceiling of nonconcessional loans.

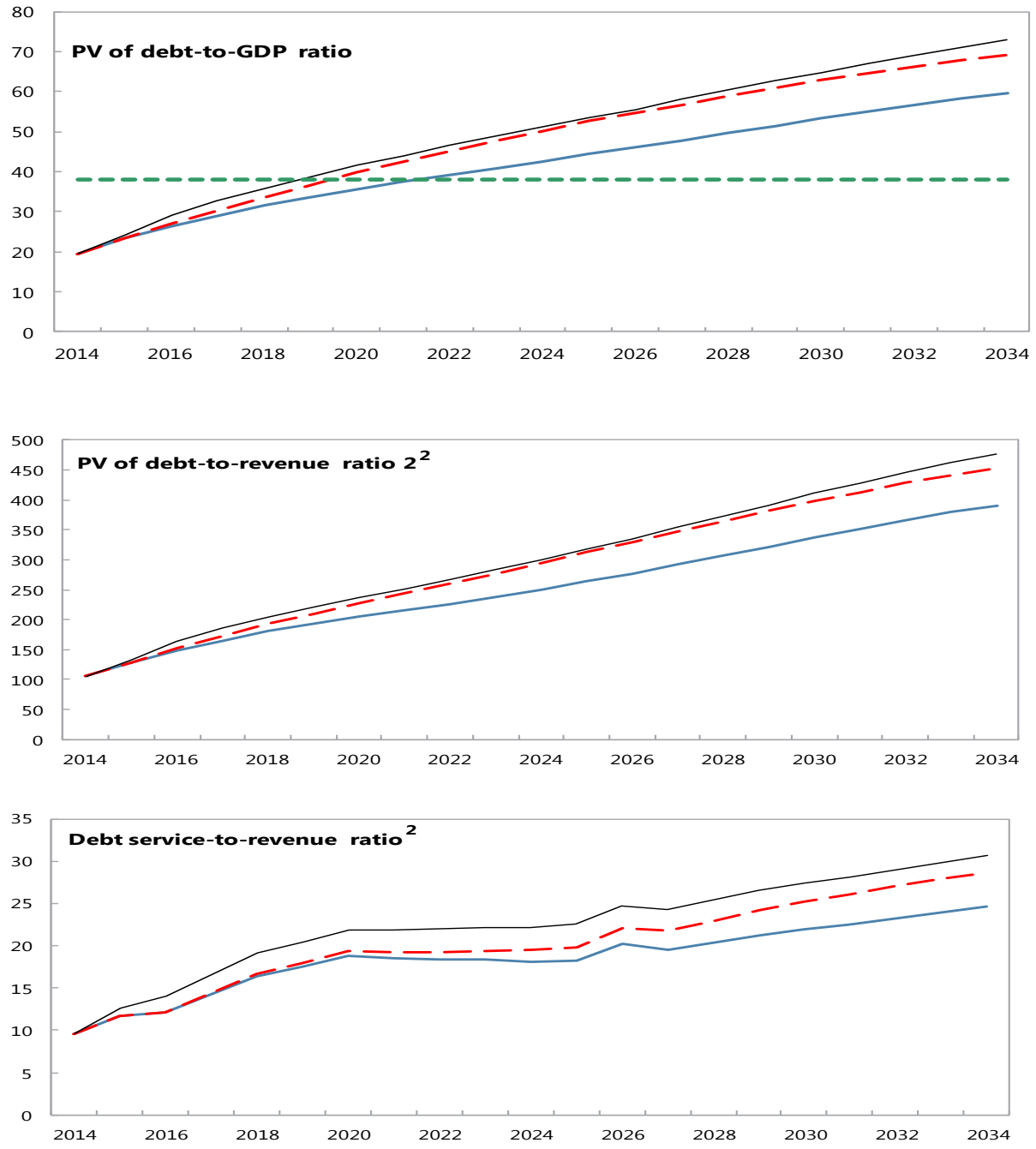
Figure 1. Cameroon: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2014–34[1]



Sources: Country authorities, and IMF and World Bank staff estimates and projections.

[1] The most extreme stress test is the test that yields the highest ratio on or before 2024. In figures b, d and f, it corresponds to a non-debt flows shock; in figures c and e, it corresponds to an exports shock.

Figure 2. Cameroon: Indicators of Public Debt Under Alternative Scenarios, 2014–34[1]



Sources: Country authorities; and IMF and World Bank staff estimates and projections.

[1] The most extreme stress test is the test that yields the highest ratio on or before 2023.

[2] Revenues are defined inclusive of grants.

Table 1. Cameroon: Public Sector Debt Sustainability, Baseline Scenario, 2011–34

(Percent of GDP, unless otherwise indicated)

	Actual					Estimate						Projections			
	2011	2012	2013	Average 5/	Standard Deviation 5/	2014	2015	2016	2017	2018	2019	2014-19 Average	2024	2034	2020-34 Average
Public sector debt 1/	13.8	16.1	19.5			24.0	28.0	31.1	33.8	36.3	38.4		48.6	68.3	
of which: foreign-currency denominated	7.7	9.3	12.5			14.7	16.4	17.2	17.7	17.8	17.8		18.0	15.8	
Change in public sector debt	1.7	2.3	3.4			4.5	4.0	3.0	2.7	2.5	2.1		2.0	1.7	
Identified debt-creating flows	1.8	1.2	2.1			4.0	3.7	2.8	2.5	2.2	1.9		1.8	1.7	
Primary deficit	2.4	1.3	3.6	-3.7	10.8	5.1	5.1	4.3	4.1	4.0	3.7	4.4	3.8	4.6	4.1
Revenue and grants	18.7	18.8	18.1			18.6	18.2	17.7	17.6	17.5	17.5		17.0	15.3	
of which: grants	0.5	0.4	0.3			0.5	0.5	0.3	0.3	0.2	0.2		0.1	0.0	
Primary (noninterest) expenditure	21.2	20.1	21.8			23.6	23.3	22.0	21.7	21.4	21.2		20.8	19.9	
Automatic debt dynamics	-0.6	-0.6	-1.7			-1.1	-1.3	-1.5	-1.6	-1.7	-1.8		-2.0	-2.9	
Contribution from interest rate/growth differential	-0.5	-0.6	-0.9			-1.0	-1.1	-1.3	-1.4	-1.6	-1.6		-2.0	-2.9	
of which: contribution from average real interest rate	0.0	0.0	-0.1			0.0	0.2	0.2	0.2	0.2	0.3		0.4	0.5	
of which: contribution from real GDP growth	-0.5	-0.6	-0.8			-1.0	-1.3	-1.5	-1.6	-1.8	-1.9		-2.4	-3.4	
Contribution from real exchange rate depreciation	-0.1	0.0	-0.8			-0.1	-0.2	-0.2	-0.2	-0.2	-0.2		...	...	
Other identified debt-creating flows	0.0	0.5	0.2			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.5	0.2			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-0.1	1.1	1.3			0.5	0.3	0.3	0.3	0.2	0.2		0.2	0.0	
Other Sustainability Indicators															
PV of public sector debt	...	...	14.7			19.5	23.3	26.4	29.3	31.9	34.2		44.7	64.8	50.9
of which: foreign-currency denominated	...	...	7.7			10.2	11.8	12.6	13.1	13.4	13.6		14.1	12.4	13.5
of which: external	...	...	7.7			10.2	11.8	12.6	13.1	13.4	13.6		14.1	12.4	13.5
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	3.7	3.0	5.1			6.9	7.2	6.4	6.6	6.8	6.8		7.0	8.6	7.6
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	80.9			105.1	128.0	148.8	166.7	182.8	195.3		262.3	424.0	313.6
PV of public sector debt-to-revenue ratio (in percent)	...	...	82.4			108.2	131.6	151.3	169.1	185.0	197.3		263.9	425.1	315.1
of which: external 3/	...	...	43.0			56.4	66.3	72.1	75.9	77.9	78.6		83.0	81.0	83.1
Debt service-to-revenue and grants ratio (in percent) 4/	7.0	8.9	8.4			9.6	11.7	12.1	14.3	16.5	17.6		18.6	26.1	21.3
Debt service-to-revenue ratio (in percent) 4/	7.2	9.1	8.5			9.9	12.0	12.3	14.5	16.7	17.8		18.7	26.1	21.4
Primary deficit that stabilizes the debt-to-GDP ratio	0.7	-1.0	0.2			0.6	1.1	1.2	1.4	1.5	1.6		1.8	2.9	2.1
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	4.1	4.6	5.2	3.5	1.0	5.5	5.5	5.5	5.5	5.5	5.5	5.5	5.5	5.4	5.5
Average nominal interest rate on forex debt (in percent)	2.8	2.4	3.2	1.8	0.7	1.9	2.3	2.4	2.6	2.7	2.8	2.5	3.1	3.3	3.2
Average real interest rate on domestic debt (in percent)	-1.6	-0.3	-4.4	-1.0	2.5	-0.1	1.0	0.7	1.0	0.9	0.9	0.7	0.7	0.7	0.7
Real exchange rate depreciation (in percent, + indicates depreciation	-1.2	-0.5	-8.7	-2.1	4.3	-1.0	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	3.7	2.8	6.3	2.7	2.6	2.0	1.8	2.0	1.9	1.9	1.9	1.9	1.8	1.8	1.8
Growth of real primary spending (deflated by GDP deflator, in percer	19.5	-0.6	13.9	3.3	7.2	14.6	4.1	-0.2	3.8	4.2	4.6	5.2	5.2	5.0	5.1
Grant element of new external borrowing (in percent)	...	...	...	...	...	21.2	20.3	19.1	18.2	17.6	17.1	18.9	16.5	15.4	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2. Cameroon: Sensitivity Analysis for Key Indicators of Public Debt, 2014–34

	Projections								
	2014	2015	2016	2017	2018	2019	2024	2025	2034
PV of debt-to-GDP ratio									
Baseline	19	23	26	29	32	34	45	47	65
A. Alternative scenarios									
A1. Real GDP growth and primary balance are at historical averages	19	18	16	14	13	11	3	2	-6
A2. Primary balance is unchanged from 2014	19	23	27	30	34	37	52	54	73
A3. Permanently lower GDP growth 1/	19	23	27	30	32	35	47	49	72
B. Bound tests									
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	19	24	29	33	36	39	53	56	78
B2. Primary balance is at historical average minus one standard deviations in 2015-201	19	25	30	32	35	37	47	49	66
B3. Combination of B1-B2 using one half standard deviation shocks	19	22	23	27	31	34	48	51	74
B4. One-time 30 percent real depreciation in 2015	19	27	30	32	35	37	46	48	66
B5. 10 percent of GDP increase in other debt-creating flows in 2015	19	30	33	36	38	40	49	51	68
PV of debt-to-revenue ratio [2]									
Baseline	105	128	149	167	183	195	262	279	424
A. Alternative scenarios									
A1. Real GDP growth and primary balance are at historical averages	105	97	90	81	73	64	18	10	-42
A2. Primary balance is unchanged from 2014	105	128	152	173	194	211	303	323	480
A3. Permanently lower GDP growth 1/	105	128	150	169	186	199	275	294	468
B. Bound tests									
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	105	134	164	187	207	224	313	333	511
B2. Primary balance is at historical average minus one standard deviations in 2015-201	105	136	167	184	200	211	276	292	433
B3. Combination of B1-B2 using one half standard deviation shocks	105	118	132	155	176	194	282	303	482
B4. One-time 30 percent real depreciation in 2015	105	149	168	184	198	209	272	288	432
B5. 10 percent of GDP increase in other debt-creating flows in 2015	105	166	186	202	217	228	290	305	442
Debt service-to-revenue ratio [2]									
Baseline	10	12	12	14	16	18	19	19	26
A. Alternative scenarios									
A1. Real GDP growth and primary balance are at historical averages	10	12	12	13	15	16	11	10	1
A2. Primary balance is unchanged from 2014	10	12	12	14	17	18	20	20	30
A3. Permanently lower GDP growth 1/	10	12	12	14	17	18	19	20	28
B. Bound tests									
B1. Real GDP growth is at historical average minus one standard deviations in 2015-20	10	12	13	15	18	19	21	21	31
B2. Primary balance is at historical average minus one standard deviations in 2015-201	10	12	12	15	17	18	20	20	27
B3. Combination of B1-B2 using one half standard deviation shocks	10	12	12	14	17	18	19	19	29
B4. One-time 30 percent real depreciation in 2015	10	13	14	17	19	20	23	23	32
B5. 10 percent of GDP increase in other debt-creating flows in 2015	10	12	13	16	17	19	21	21	28

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.

Table 3a. Cameroon: External Debt Sustainablity Framework, Baseline Scenario, 2010–34

(Percent of GDP, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2011	2012	2013			2014	2015	2016	2017	2018	2019	2014-2019 Average	2024	2034	2020-2034 Average
External debt (nominal) 1/	**7.7**	**9.3**	**12.5**			**14.7**	**16.4**	**17.2**	**17.7**	**17.8**	**17.8**		**18.0**	**15.8**	
of which: public and publicly guaranteed (PPG)	7.7	9.3	12.5			14.7	16.4	17.2	17.7	17.8	17.8		18.0	15.8	
Change in external debt	1.2	1.6	3.3			2.2	1.8	0.8	0.4	0.1	0.0		-0.1	-0.4	
Identified net debt-creating flows	0.1	2.0	2.1			2.8	2.5	1.3	0.9	0.6	0.4		-0.5	-0.4	
Non-interest current account deficit	**2.8**	**3.8**	**3.8**	**1.6**	**1.5**	**3.6**	**3.9**	**4.0**	**4.0**	**4.1**	**4.0**		**3.6**	**3.0**	3.6
Deficit in balance of goods and services	2.6	3.0	3.3			3.0	3.3	3.4	3.5	3.5	3.5		2.9	2.3	2.9
Exports	29.6	29.2	27.2			27.0	26.4	25.7	25.4	25.1	25.0		24.8	23.3	24.2
Imports	32.2	32.2	30.4			30.0	29.8	29.2	28.8	28.7	28.5		27.7	25.6	
Net current transfers (negative = inflow)	-1.0	-0.8	-1.1	-1.5	0.6	-1.1	-1.0	-1.0	-0.9	-0.9	-0.8		-0.6	-0.3	-0.5
of which: official	-1.0	-0.9	-0.3			-0.3	-0.3	-0.3	-0.2	-0.2	-0.2		-0.2	-0.1	
Other current account flows (negative = net inflow)	1.1	1.6	1.6			1.6	1.6	1.5	1.5	1.4	1.4		1.3	0.9	
Net FDI (negative = inflow)	**-2.1**	**-2.1**	**-0.7**	**-4.2**	**7.9**	**-0.4**	**-1.0**	**-2.3**	**-2.6**	**-3.1**	**-3.1**		**-3.7**	**-3.1**	-3.6
Endogenous debt dynamics 2/	**-0.6**	**0.2**	**-1.0**			**-0.4**	**-0.4**	**-0.5**	**-0.5**	**-0.5**	**-0.4**		**-0.4**	**-0.3**	
Contribution from nominal interest rate	0.2	0.2	0.3			0.2	0.3	0.4	0.4	0.4	0.5		0.5	0.5	
Contribution from real GDP growth	-0.2	-0.4	-0.4			-0.6	-0.7	-0.8	-0.9	-0.9	-0.9		-0.9	-0.8	
Contribution from price and exchange rate changes	-0.5	0.4	-0.8			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**1.1**	**-0.4**	**1.2**			**-0.6**	**-0.7**	**-0.5**	**-0.5**	**-0.4**	**-0.4**		**0.4**	**0.0**	
of which: exceptional financing	-1.3	0.3	-0.3			-0.1	-0.1	-0.1	-0.1	-0.1	-0.1		0.0	0.0	
PV of external debt 4/	...	...	7.7			10.2	11.8	12.6	13.1	13.4	13.6		14.1	12.4	
In percent of exports	...	...	28.2			37.7	44.4	48.9	51.8	53.5	54.3		56.6	53.1	
PV of PPG external debt	**...**	**...**	**7.7**			**10.2**	**11.8**	**12.6**	**13.1**	**13.4**	**13.6**		**14.1**	**12.4**	
In percent of exports	**...**	**...**	**28.2**			**37.7**	**44.4**	**48.9**	**51.8**	**53.5**	**54.3**		**56.6**	**53.1**	
In percent of government revenues	**...**	**...**	**43.0**			**56.4**	**66.3**	**72.1**	**75.9**	**77.9**	**78.6**		**83.0**	**81.0**	
Debt service-to-exports ratio (in percent)	**1.1**	**1.7**	**1.9**			**2.7**	**3.1**	**3.3**	**3.6**	**4.3**	**4.5**		**5.9**	**6.6**	
PPG debt service-to-exports ratio (in percent)	**1.1**	**1.7**	**1.9**			**2.7**	**3.1**	**3.3**	**3.6**	**4.3**	**4.5**		**5.9**	**6.6**	6.2
PPG debt service-to-revenue ratio (in percent)	**1.7**	**2.6**	**2.8**			**4.0**	**4.6**	**4.8**	**5.3**	**6.2**	**6.5**		**8.7**	**10.1**	9.2
Total gross financing need (Billions of U.S. dollars)	0.3	0.6	1.0			1.3	1.3	1.0	1.0	1.0	1.0		1.0	2.2	1.3
Non-interest current account deficit that stabilizes debt ratio	1.6	2.3	0.5			1.4	2.1	3.2	3.6	4.0	3.9		3.6	3.3	3.7
Key macroeconomic assumptions															
Real GDP growth (in percent)	4.1	4.6	5.2	3.5	1.0	5.5	5.5	5.5	5.5	5.5	5.5	5.5	5.5	5.4	5.5
GDP deflator in US dollar terms (change in percent)	8.7	-5.0	9.8	4.6	7.4	5.1	3.7	3.2	3.5	3.3	3.3	3.7	1.8	1.8	1.8
Effective interest rate (percent) 5/	2.8	2.4	3.2	1.8	0.7	1.9	2.3	2.4	2.6	2.7	2.8	2.5	3.1	3.3	3.2
Growth of exports of G&S (US dollar terms, in percent)	32.4	-2.0	7.6	10.9	18.0	10.0	7.3	6.0	7.7	7.9	8.8	7.9	6.7	7.2	6.9
Growth of imports of G&S (US dollar terms, in percent)	26.5	-0.7	9.2	11.8	14.8	9.3	8.6	6.7	7.9	8.4	8.5	8.2	6.5	6.5	6.7
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	21.2	20.3	19.1	18.2	17.6	17.1	18.9	16.5	15.4	16.1
Government revenues (excluding grants, in percent of GDP)	18.2	18.3	17.8			18.0	17.7	17.5	17.3	17.2	17.3		16.9	15.3	16.3
Aid flows (in Billions of US dollars) 7/	0.4	0.5	0.4			0.5	0.5	0.5	0.6	0.6	0.6		0.8	0.1	
of which: Grants	0.1	0.1	0.1			0.2	0.2	0.1	0.1	0.1	0.1		0.1	0.1	
of which: Concessional loans	0.2	0.4	0.3			0.4	0.4	0.4	0.5	0.5	0.5		0.7	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			1.3	1.2	0.8	0.7	0.6	0.6		0.5	0.3	0.4
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			31.2	30.2	27.3	26.0	24.8	23.4		20.5	17.2	19.3
Memorandum items:															
Nominal GDP (Billions of US dollars)	25.5	25.3	29.3			32.5	35.5	38.7	42.3	46.1	50.3		72.1	148.1	
Nominal dollar GDP growth	13.2	-0.7	15.6			10.9	9.4	8.9	9.2	9.1	9.0	9.4	7.4	7.3	7.5
PV of PPG external debt (in Billions of US dollars)			2.3			3.3	4.2	4.9	5.6	6.2	6.9		10.2	18.4	
(PVt-PVt-1)/GDPt-1 (in percent)						3.5	2.7	2.0	1.8	1.5	1.4	2.1	1.0	0.6	0.9
Gross workers' remittances (Billions of US dollars)	0.4	0.3	0.5			0.5	0.5	0.6	0.6	0.6	0.6		0.7	0.8	
PV of PPG external debt (in percent of GDP + remittances)	...	...	7.5			10.0	11.6	12.4	13.0	13.3	13.4		13.9	12.3	
PV of PPG external debt (in percent of exports + remittances)	...	...	26.5			35.6	42.0	46.3	49.2	50.8	51.8		54.5	51.9	
Debt service of PPG external debt (in percent of exports + remittance	...	...	1.7			2.5	2.9	3.1	3.4	4.1	4.3		5.7	6.5	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 3b. Cameroon: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–34

(Percent)

	Projections							
	2014	2015	2016	2017	2018	2019	**2024**	2034
PV of debt-to GDP ratio								
Baseline	10	12	13	13	13	14	**14**	12
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	10	8	6	4	2	1	**-5**	-8
A2. New public sector loans on less favorable terms in 2014-2034 2	10	12	13	14	15	15	**16**	16
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	10	12	13	14	14	15	**15**	13
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	10	14	20	20	20	20	**18**	13
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	10	13	14	15	15	16	**16**	14
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	10	15	20	20	20	20	**19**	14
B5. Combination of B1-B4 using one-half standard deviation shocks	10	14	19	19	19	19	**19**	14
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	10	17	18	19	19	19	**20**	18
PV of debt-to-exports ratio								
Baseline	38	44	49	52	53	54	**57**	53
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	38	30	23	16	9	3	**-21**	-35
A2. New public sector loans on less favorable terms in 2014-2034 2	38	46	52	56	58	59	**65**	67
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	38	45	49	52	54	55	**57**	53
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	38	63	100	103	104	104	**98**	76
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	38	45	49	52	54	55	**57**	53
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	38	58	79	81	81	81	**76**	58
B5. Combination of B1-B4 using one-half standard deviation shocks	38	53	72	75	76	76	**75**	62
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	38	45	49	52	54	55	**57**	53
PV of debt-to-revenue ratio								
Baseline	56	66	72	76	78	79	**83**	81
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	56	45	35	24	14	4	**-30**	-54
A2. New public sector loans on less favorable terms in 2014-2034 2	56	69	77	82	84	86	**95**	102
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	56	69	77	81	83	84	**89**	86
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	56	81	112	114	115	113	**109**	88
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	56	71	82	87	89	90	**95**	92
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	56	86	116	118	118	117	**112**	89
B5. Combination of B1-B4 using one-half standard deviation shocks	56	79	108	111	112	111	**110**	95
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	56	94	102	108	110	111	**118**	115

Table 3b. Cameroon: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2014–34 (concluded)

(Percent)

Debt service-to-exports ratio								
Baseline	3	3	3	4	4	4	**6**	7
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	3	3	3	3	3	3	**2**	0
A2. New public sector loans on less favorable terms in 2014-2034 2	3	3	3	3	3	3	**4**	5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	3	3	3	4	4	4	**6**	7
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	3	4	5	6	7	7	**10**	10
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	3	3	3	4	4	4	**6**	7
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	3	3	4	4	5	5	**8**	8
B5. Combination of B1-B4 using one-half standard deviation shocks	3	3	4	4	5	5	**8**	8
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	3	3	3	4	4	4	**6**	7
Debt service-to-revenue ratio								
Baseline	4	5	5	5	6	6	**9**	10
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2014-2034 1/	4	5	4	4	5	5	**3**	0
A2. New public sector loans on less favorable terms in 2014-2034 2	4	5	4	4	5	5	**6**	8
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2015-2016	4	5	5	6	7	7	**9**	11
B2. Export value growth at historical average minus one standard deviation in 2015-2016 3/	4	5	5	6	7	7	**11**	12
B3. US dollar GDP deflator at historical average minus one standard deviation in 2015-2016	4	5	5	6	7	7	**10**	11
B4. Net non-debt creating flows at historical average minus one standard deviation in 2015-2016 4/	4	5	5	6	7	7	**12**	12
B5. Combination of B1-B4 using one-half standard deviation shocks	4	5	6	7	8	8	**11**	12
B6. One-time 30 percent nominal depreciation relative to the baseline in 2015 5/	4	7	7	7	9	9	**12**	14
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	29	29	29	29	29	29	**29**	29

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the b
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after th an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.